Design and Realization of an Affordable Continuous Sub-Cutaneous Insulin Infusion Pump for Precise Insulin Delivery

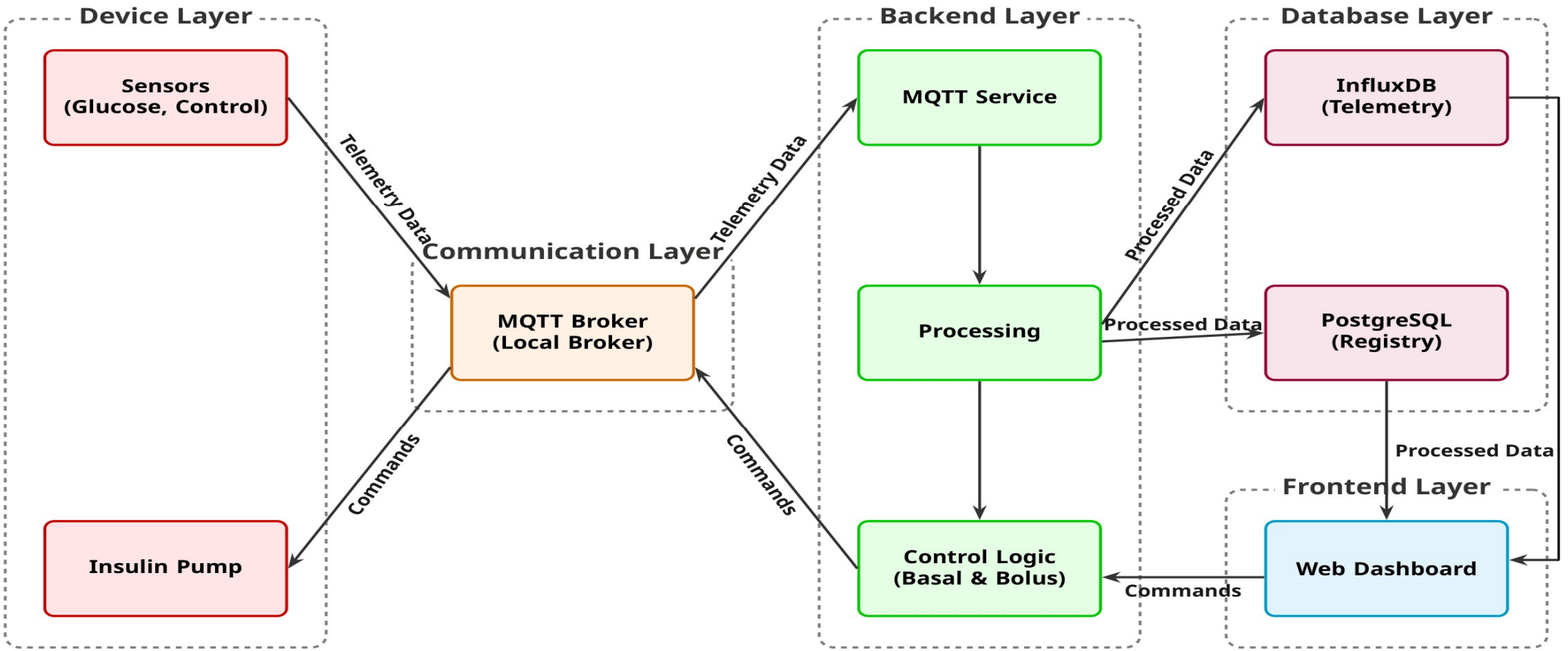
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Project Mentor: Dr. Anirudh Nath



WebApp Design



WebApp Design

Closed-Loop Insulin Infusion Pump Data Flow Overview :

1. Data Collection and Transmission (Device → Backend)

Continuous Monitoring

Sensors continuously track physiological parameters
Includes glucose levels and device status

Telemetry Data Transmission

Data is transmitted using the MQTT over local network
Direct communication between device and web app
Ensures low-latency and reliable message exchange

Backend Ingestion

MQTT Service subscribes to incoming data
Telemetry is received and forwarded for processing

2. Data Processing and Storage (Backend → Database)

Data Processing

Raw data is cleaned and formatted
Processing layer prepares data for storage and analysis

Time-Series Storage (InfluxDB)

Stores continuous sensor data over time
Optimized for high-frequency telemetry

Relational Storage (PostgreSQL)

Stores structured data such as:
- Device registry
- Patient information & System state logs

WebApp Design

Closed-Loop Insulin Infusion Pump Data Flow Overview :

3. Pump Control (Backend → Pump)

Command Generation and Dispatch

- Control Logic generates machine-level commands
- Commands are sent to the MQTT Broker for delivery

Insulin Delivery

- Insulin Pump subscribes to its command topic
- Receives instructions from MQTT
- Adjusts insulin infusion rate accordingly

4. User Monitoring and Intervention

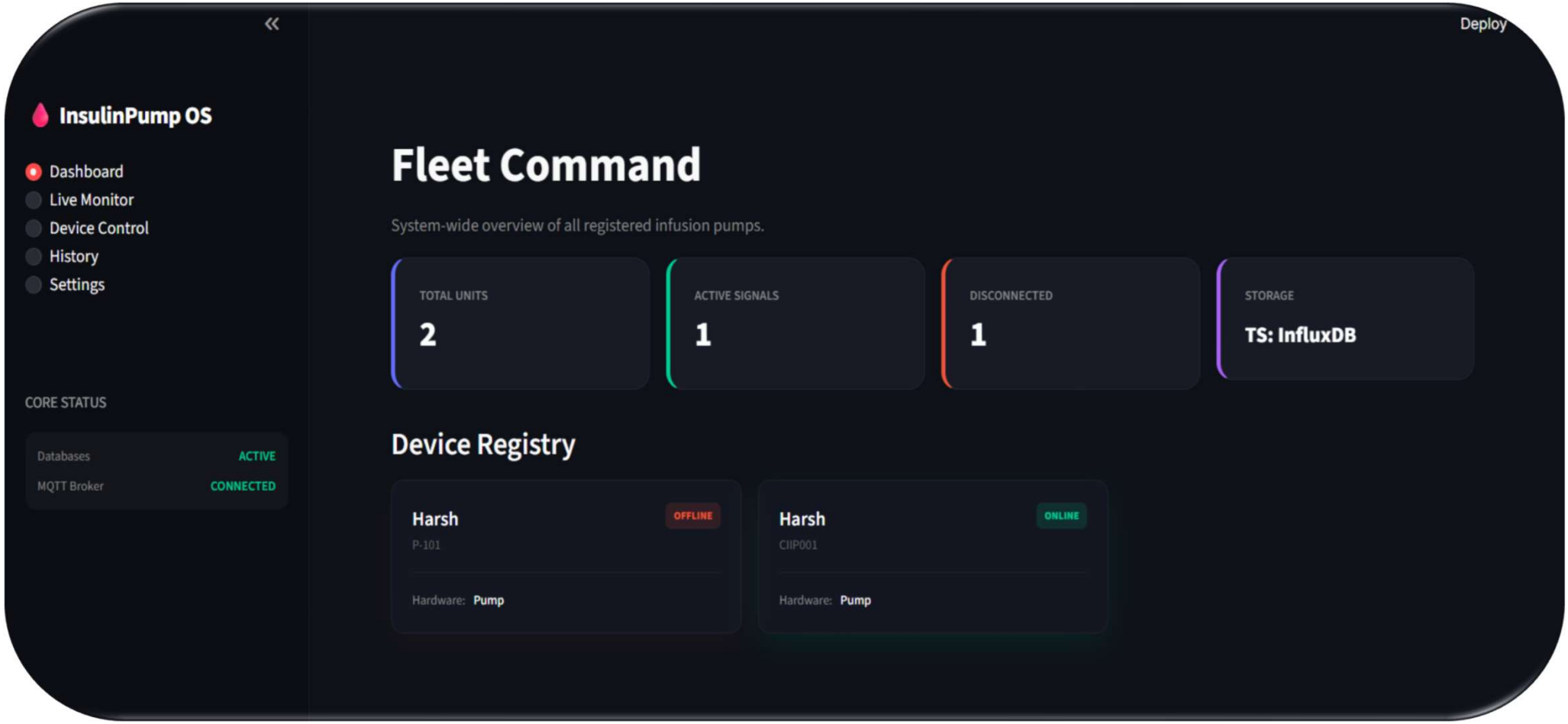
Real-Time Dashboard

- Displays processed
- Visualizes glucose levels, pump status, and history

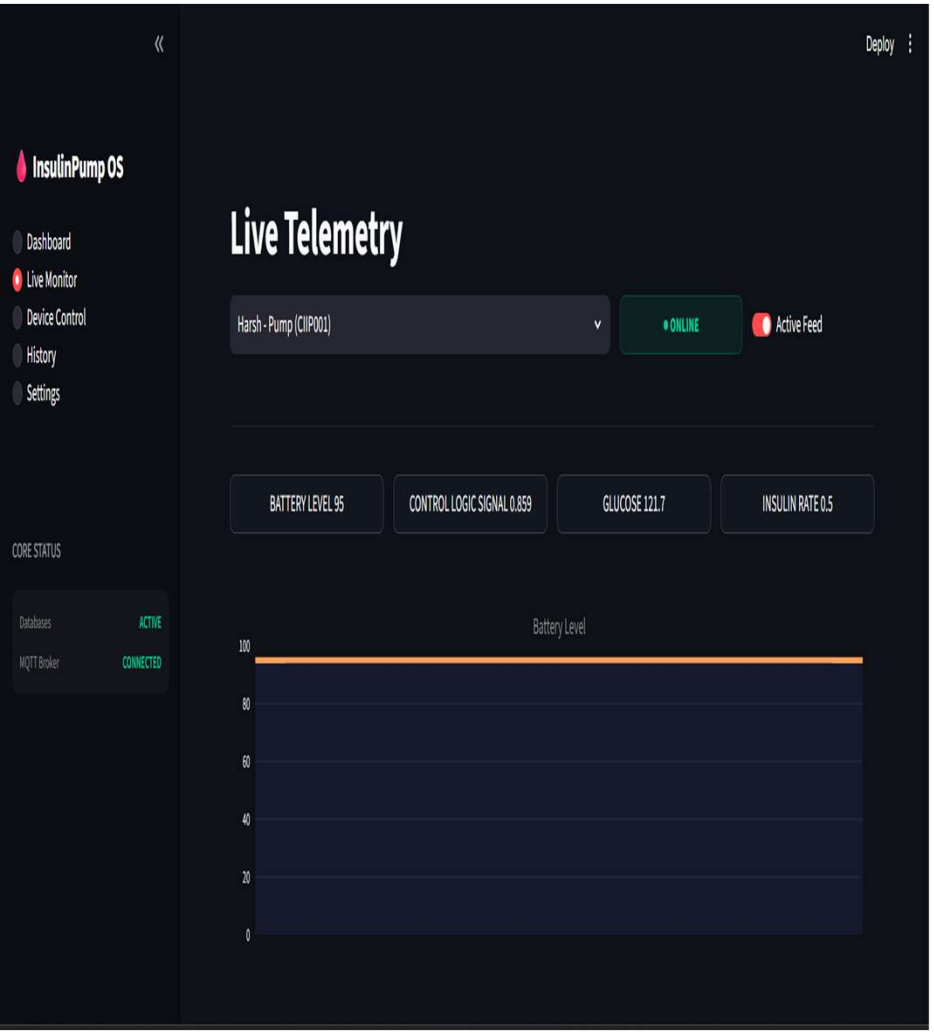
Manual Control

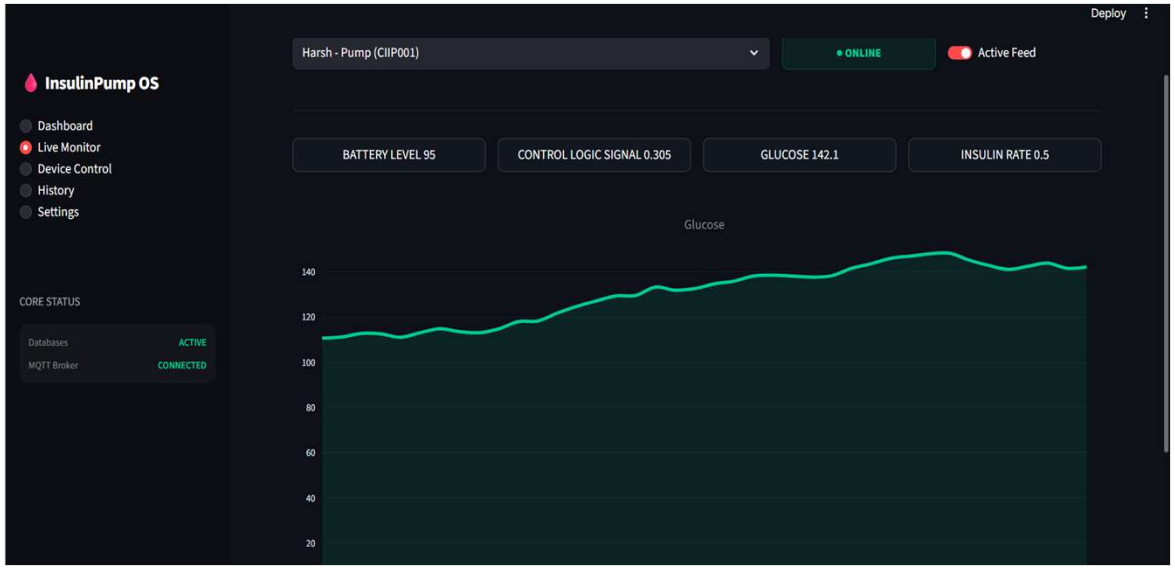
- Users can input commands (e.g., bolus dose, settings adjustment)
- Enables human intervention when required

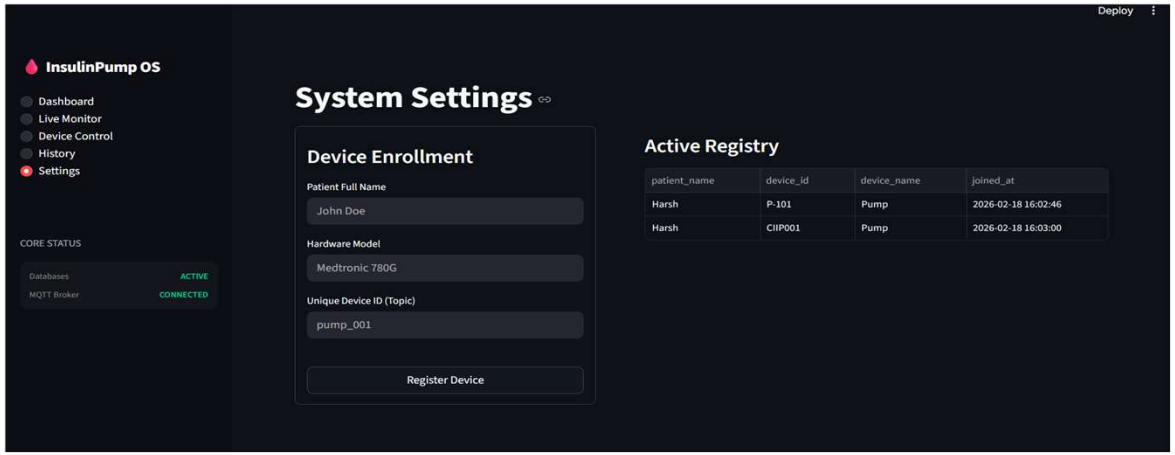
Preview


Deploy
InsulinPump OS
Dashboard
Live Monitor
Device Control
History
Settings
CORE STATUS
Databases
ACTIVE
MQTT Broker
CONNECTED
Fleet Command
System-wide overview of all registered infusion pumps.
TOTAL UNITS
2
ACTIVE SIGNALS
1
DISCONNECTED
1
STORAGE
TS: InfluxDB
Device Registry
Harsh
OFFLINE
P-101
Hardware: Pump
Harsh
ONLINE
CIIP001
Hardware: Pump

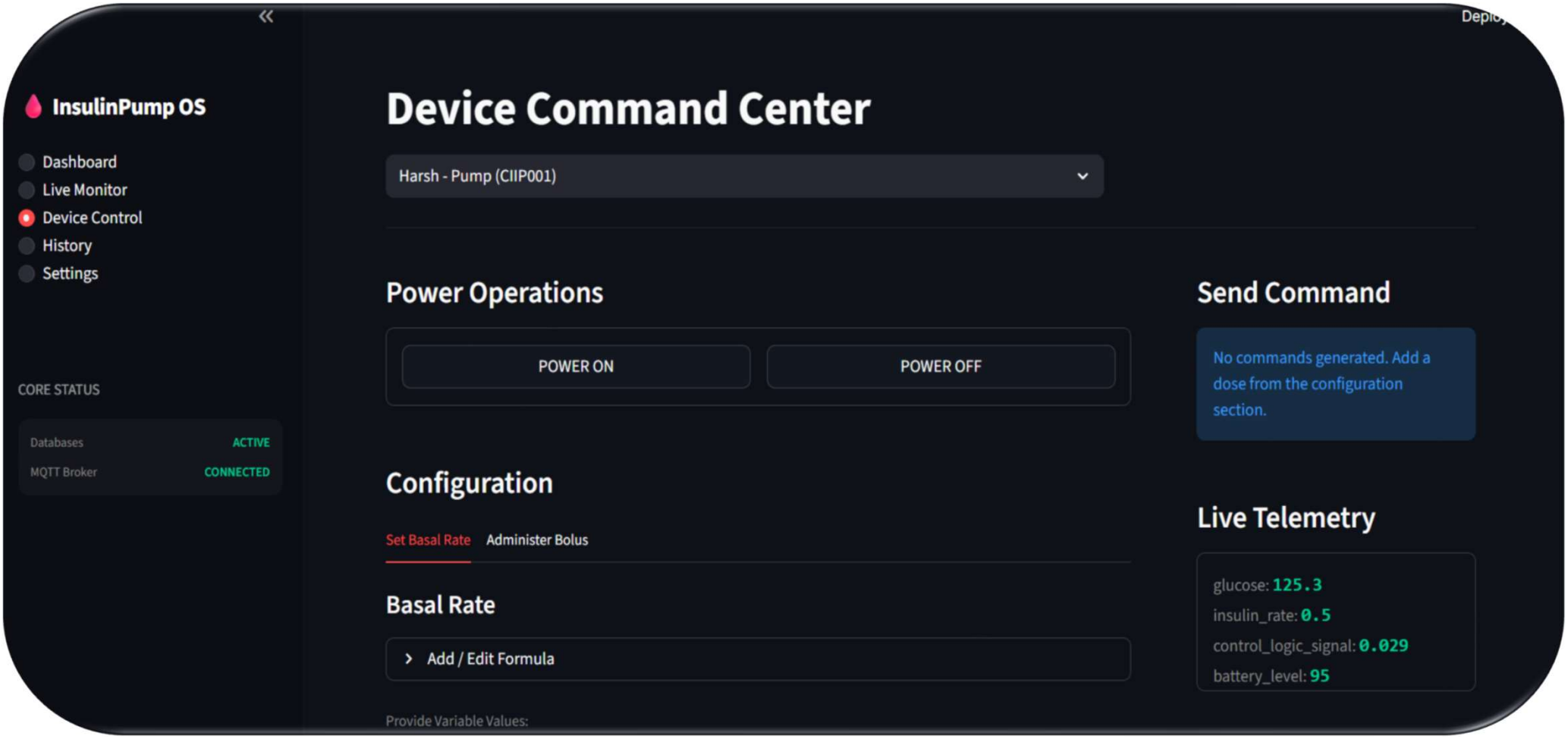
Preview


Deploy
InsulinPump OS
Dashboard
Live Monitor
Device Control
History
Settings
Live Telemetry
Harsh - Pump (CIIP001)
ONLINE
Active Feed
BATTERY LEVEL 95
CONTROL LOGIC SIGNAL 0.859
GLUCOSE 121.7
INSULIN RATE 0.5
CORE STATUS
Databases
ACTIVE
MQTT Broker
CONNECTED
Battery Level
100
80
60
40
20
0


Deploy
InsulinPump OS
Dashboard
Live Monitor
Device Control
History
Settings
CORE STATUS
Databases
ACTIVE
MQTT Broker
CONNECTED
Harsh - Pump (CIIP001)
ONLINE
Active Feed
BATTERY LEVEL 95
CONTROL LOGIC SIGNAL 0.305
GLUCOSE 142.1
INSULIN RATE 0.5
Glucose
140
120
100
80
60
40
20


Deploy
InsulinPump OS
Dashboard
Live Monitor
Device Control
History
Settings
CORE STATUS
Databases
ACTIVE
MQTT Broker
CONNECTED
System Settings
Device Enrollment
Patient Full Name
John Doe
Hardware Model
Medtronic 780G
Unique Device ID (Topic)
pump_001
Register Device
Active Registry
patient_name
device_id
device_name
joined_at
Harsh
P-101
Pump
2026-02-18 16:02:46
Harsh
CIIP001
Pump
2026-02-18 16:03:00

Preview


InsulinPump OS
Dashboard
Live Monitor
Device Control
History
Settings
CORE STATUS
Databases
ACTIVE
MQTT Broker
CONNECTED
Device Command Center
Harsh - Pump (CIIP001)
Power Operations
POWER ON
POWER OFF
Configuration
Set Basal Rate
Administer Bolus
Basal Rate
Add / Edit Formula
Provide Variable Values:
Send Command
No commands generated. Add a dose from the configuration section.
Live Telemetry
glucose: 125.3
insulin_rate: 0.5
control_logic_signal: 0.029
battery_level: 95